

Ana C. Huamán Quispe

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Research Interests

My research focuses on generalizable robot manipulation targeting space applications, specifically: (a) creating algorithms that allow robots to operate safely without constant human supervision, and (b) developing software tools that facilitate the integration of robot and flight software for non-expert users.

Current position

Research Scientist

TRAC Labs – Houston, TX

2017 - Present

Education

PhD, Robotics

Georgia Institute of Technology, USA

Advisors: Henrik Christensen and Mike Stilman. “Grasp selection strategies for robot manipulation using a superquadric-based object representation”

2010 – 2016

MS, Computational Perception and Robotics

Georgia Institute of Technology, USA

2010 - 2013

B.Sc, Mechatronics

Universidad Nacional de Ingeniería, Perú

2003 - 2008

Selected Publications

Integrating ROS2 with the Flight Core System

A. Huamán Quispe, T. Milam, A. Harris, D. Edell

18th International Conference on Space Operations (SpaceOps), May 2025

Generalized Affordance Templates for Mobile Manipulation

S. Hart, A. Huamán Quispe, M.W. Lanighan, and S. Gee

IEEE Intl. Conference on Robotics and Automation (ICRA), 2022

PLUMMRS: A Collection of Plan Ledgers and Unified Maps for Multi-Robot Safety

A. Huamán Quispe, S. Hart, and S. Gee

16th Intl. Conference on Space Operations (SpaceOps), 2021

Learning user preferences for robot-human handovers

A. Huamán Quispe, E. Martinson, and K. Oguchi

IEEE/RSJ Intl. Conference on Intelligent Robots and Systems (IROS), 2017

Combining arm and hand metrics for sensible grasp selection

A. Huamán Quispe, H. Ben Amor, and H.I. Christensen

Conference on Automation Science and Engineering (CASE), 2016

A Taxonomy of Benchmark Tasks for Robot Manipulation

A. Huamán Quispe, H. Ben Amor and H.I. Christensen

International Symposium on Robotics Research (ISRR 2015)

Efficient Manipulation Planning with basic primitives

A. Huamán Quispe, B. Melville, C. Erdogan, H. Ben Amor, H.I. Christensen and M. Stilman

Proceedings of 2015 IEEE International Conference on Robotics and Automation (ICRA)

Handover Planning for Every Occasion

A. Huamán Quispe, Heni Ben Amor and M. Stilman

Proceedings of 2014 IEEE-RSJ International Conference on Humanoid Robots (Humanoids)

Generation of Diverse Paths in 3D Environments

A. Huamán Quispe, T. Kunz and M. Stilman

Proceedings of 2013 IEEE-RSJ International Conference on Intelligent Robots and Systems (IROS)

Deterministic Motion Planning for Redundant Robots along End-Effector Paths

A. Huamán Quispe and M. Stilman

Proceedings of 2012 IEEE-RAS International Conference on Humanoid Robots (Humanoids)

Selected Projects

BUILDER: Leveraging BIM information to automate robot task planning for lunar assembly applications

September 2025 - March 2026

NASA SBIR Phase I, Kennedy Space Center. Contract: 80NSSC25C0383.

The BRASH Integration Toolkit for ROS2 and Flight Software Interoperability

April 2022 - June 2024

NASA STTR Phase II, Ames Research Center. Contract: 80NSSC22CA020.

PLUMMRS: A Collection of Plan Ledgers and Unified Maps for Multi-Robot Safety

August 2021 - July 2024

NASA SBIR Phase II, Ames Research Center. Contract: 80NSSC21C0493

Miscellaneous

Programming and Software: C++, Python, OpenCV, PyTorch, ROS2, Mujoco, ISAAC Lab, Blender.

Nationality: Peruvian

Languages: Spanish (native), English (fluent).